Key ideas and chapter summary

Ratio

A ratio is used to compare amounts of the same units in a definite order. Equivalent ratios are obtained by multiplying or dividing by the same number.

Dividing a quantity in a given ratio

1. Convert the ratio into two fractions.
2. Multiply the quantity by both fractions.

Scale drawing

Scale of a drawing = Drawing length : Actual length
Scale is expressed in two ways:

- using units, such as 1 cm to 1 m (or 1 cm = 1 m)
- with no units, using a ratio such as 1 : 100.

Plans and elevations

A plan is a view of an object from the top. An elevation is a view of an object from one side such as a front elevation or side elevation.

Perimeter

When calculating the perimeter of the land, multiply side lengths on an aerial photograph by the scale.

Area

To estimate the area of land, divide it into square grids and count the number of squares. To calculate the area of land, use the appropriate formula.
The trapezoidal rule for an area of land with one irregular boundary gives an approximation.

Volume of water and the trapezoidal rule

The volume of a body of water of constant depth and a trapezoidal area, or volume of rainfall falling on a trapezoidal area, can be calculated using the trapezoidal rule for area, $A \approx \frac{b}{2}\left(d_f + d_l\right)$, and the formula for the volume of a right prism, $V = Ah$.

Rate

A rate is a comparison of quantities expressed in different units, such as km/h, L/100 km and $/hour.

Unitary method

Use division to go down to 1. Multiply to find the required amount.

Energy rate

Energy is the rate of power per unit of time (kilowatt hour - kWh)
Running cost of the appliance is calculated by multiplying the energy consumption by the electricity charge.

Fuel consumption

A rate, expressed in litres/100 km or km/litre.

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Review

Chapter checklist with success criteria

Check-list *A printable version of this checklist is available in the Interactive Textbook.* ☑

1A **1** I can change a ratio to a simpler form, in three ways. ☐

Consider the ratio 4 : 6.

a Express the ratio 4 : 6 in simplest form.

b Express the ratio 4 : 6 in the form 1 : a.

c Express the ratio 4 : 6 in the form b : 1.

1A **2** I can use a ratio to find an unknown quantity. ☐

If 4 : 3 is equal to a : 24, what is the value of a?

1B **3** I can use the capture-recapture technique to estimate a population. ☐

A biologist traps 50 carp from a dam, then tags and releases them. One month later he captures 40 carp, of which 16 are tagged. Estimate the number of carp in the dam.

1C **4** I can divide a quantity in a given ratio. ☐

Divide $400 in the ratio:

a 3 : 2 **b** 4 : 1 **c** 3 : 7.

1D **5** I can use a scale drawing to calculate real distances. ☐

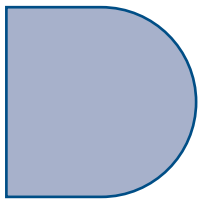
On a map the scale is 1 cm = 50 m. Two points are 3.2 cm apart on the map. How far apart are they in real life?

1E **6** I can use a scale on a building plan. ☐

A building plan has been drawn using the scale 1 : 100. Therefore, every mm on the plan represents what distance, in centimetres?

1F **7** I can calculate perimeter, area and volume from a plan and an aerial photo, using the trapezoidal rule when necessary. ☐

The following shape represents the surface of a concrete pond used for cultivating fish. It has uniform depth of 1.5 metres.



The scale is 1 : 500.

a Estimate the perimeter, in metres, by assuming the curve is a semicircle.

b Estimate the area, in square metres, using the two applications of the trapezoidal rule.

c Estimate the number of litres of water required to fill the pond.

d Judging by the shape, is this an underestimate or overestimate of the volume?

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1G **8** I can use the unitary method to solve problems involving rates. ☐

A cake for 6 people requires 4 eggs. How many eggs will be required for a cake big enough to serve 15 people?

1G **9** I can determine which product is the best buy. ☐

A drink comes in two sizes:

- 1.25 litres for $3.25
- 2 litres for $4.20.

Which is the best buy?

1G **10** I can solve problems involving speed, distance and time. ☐

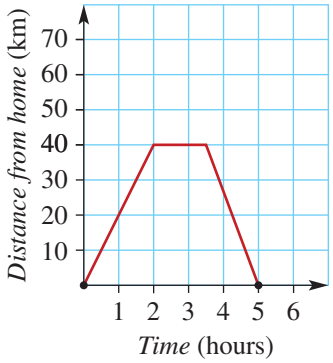
Cathy Freeman's best time for the 400-metre race was 48.63 seconds. A wombat can run at a speed of 40 km/h over short distances. If the wombat could keep up this speed for 400 metres, would it beat Cathy Freeman?

1G **11** I can convert a rate to other units of measurement. ☐

A dripping tap wastes 25 millilitres of water every minute. Convert this rate to litres per hour.

1H **12** I can interpret and analyse a distance-time graph. ☐

Consider this distance-time graph.



a What was the average speed on the first part of the journey?

b What was the average speed on the last part of the journey, correct to one decimal place?

c What was the average speed for the entire journey, including the rest break?

d What was the average speed for the entire journey, without the rest break, correct to one decimal place?

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Review

1I **13** I can calculate the cost of running electrical appliances.

The power rating for electrical appliances is shown in the table.

Electrical appliance	Power rating
Electric shaver	15 W
Hair dryer	1000 W
Iron	900 W
Laptop computer	50 W
Microwave	1200 W
Vacuum cleaner	600 W

a How much energy does a laptop computer use for 12 hours? Answer in kilowatt-hours.

b How much energy does a microwave use for $\frac{1}{2}$ hour? Answer in watt-hours.

c For how many hours was the vacuum cleaner used, if the cost of operating the vacuum cleaner was \$6.36? The price of electricity is \$0.2650 per kilowatt-hour.

1J **14** I can calculate the fuel consumption rate of a motor vehicle.

Rata's car uses 11.26 litres per 100 km.

a How many litres of petrol will his car use on a trip of 155 km from Sydney to Newcastle? Answer correct to three decimal places.

b The cost of petrol is \$1.60 per litre. How much will the petrol cost for the 155 km trip?

c How many litres of petrol will his car use on a trip of 294 km from Armidale to Taree? Answer correct to three decimal places.

d The cost of petrol is \$1.35 per litre. How much will the petrol cost for the 294 km trip?

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Multiple-choice questions

1A **1** The ratio of adults to children in a park is 5 : 9. How many adults are in the park if there are 630 children?

A 70 **B** 126
C 280 **D** 350

1C **2** A 360-gram lolly bag is divided in the ratio 7 : 5. What is the mass of the smaller amount?

A 150 g **B** 168 g
C 192 g **D** 210 g

1D **3** A scale drawing has a scale of 1 : 20. What is the actual length if the drawing length of an object is 20 mm?

A 1 mm **B** 20 mm
C 40 mm **D** 400 mm

1D **4** The scale on a map is given as 1 mm = 150 m. If the distance between two points is 600 m, what is the map distance between these points?

A 4 mm **B** 0.25 mm
C 2.5 cm **D** 40 cm

1G **5** Youssef is a courier who delivers 1 parcel on average every 25 minutes. How many hours does it take to deliver 24 parcels?

A 10 **B** 11
C 12 **D** 13

1G **6** A hose fills a 10 L bucket in 20 seconds. What is the rate of flow in litres per hour?

A 0.0001 **B** 30
C 1800 **D** 7200

1G **7** Which of the following is the slowest speed?

A 60 km/h **B** 100 m/s
C 10 000 m/min **D** 6000 m/h

1I **8** How much energy does a 600-watt hair dryer use in 5 hours?

A 1.2 kWh **B** 3 kWh
C 120 kWh **D** 3000 kWh

1I **9** Flynn uses a 1.2-kilowatt dishwasher for 5 hours. He pays 25.72 cents per kilowatt-hour. What is the cost of this usage?

A \$0.31 **B** \$1.29
C \$1.54 **D** \$6.00

1J **10** What is the fuel consumption for a vehicle that travelled 340 km using 51 litres of petrol?

A 7 L/100 km **B** 9 L/100 km
C 15 L/100 km **D** 17 L/100 km

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Review

Short-answer questions

1A **1** Simplify the following ratios.

a 500 : 100 **b** 20 : 30 **c** 28 : 7 **d** 10 : 15 : 30

e 12 : 9 **f** 56 : 88 **g** 4.8 : 1.6 **h** $\frac{3}{4} : \frac{1}{2}$

1A **2** Gabriel mixes sand and cement in the ratio 5 : 2 by volume. If he uses 5 buckets of cement, how much sand should he use?



1A **3** The ratio of Victoria's height to Willow's is 8 : 7. If Victoria is 176 cm tall, how tall is Willow?



1A **4** Express each of the following scales as a ratio in the form 1 : x.

a 1 cm to 3 m **b** 1 mm to 6 cm **c** 1 m to 2.5 km

1B **5** A scientist catches 200 fish from a pond, tags them and releases them. One month later, they catch 50 fish, of which 20 have tags. Estimate the number of fish in the pond.

1C **6** Daniel and Eddie own a business and share the profits in the ratio 3 : 4.

a The profit last week was $3437. How much does Daniel receive?

b The profit this week is $2464. How much does Eddie receive?

1D **7** The scale on a map is given as 1 cm = 5 km. If the distance between two points on the map is 46 mm, what is the actual distance between these points? Answer in kilometres.

1D **8** A scale drawing has a scale of 1 : 50 000. What is the drawing length of these actual lengths? Express your answer in millimeters.

a 4 km **b** 1250 m

c 5000 cm **d** 6.5 km

e 20 000 mm **f** 2125 m

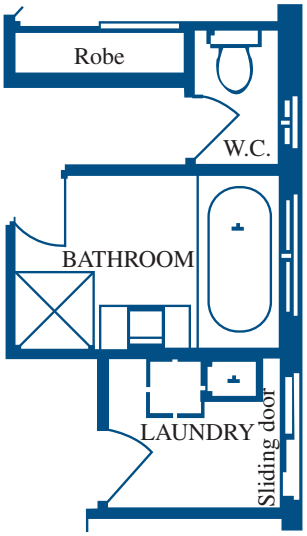
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1D **9** The scale on a map is 1 : 400. Calculate the actual distances if the following are the distances on the map. Express your answer in metres.

a Bike path 180 cm **b** Town centre 20 cm **c** Street 5 cm
d Beach 210 mm **e** River 62 cm **f** Park 60 mm

1E **10** A section of a floor plan is shown opposite.
The longer dimension of the laundry is 3 metres.

a Estimate a suitable scale for the floor plan.
b What symbol is used for the bath?
c What does 'W.C.' represent on the plan?
d What are the dimensions of the laundry?



1F **11** A body of water has the following dimensions: $b = 6$ m, $d_f = 20$ m, $d_l = 40$ m and $h = 2.5$ m. Use the trapezoidal rule for area and the formula for the volume of a right prism to answer the following.

a Find the approximate volume of the body of water.
b Find the approximate volume of water that falls on the same area when there is 9 mm of rain.

1G **12** Convert each rate to the units indicated.

a \$15/kg to \$/g **b** 14 400 m/h to m/ min
c 120 cm/s to mm/min **d** 4800 kg/g to kg/mg
e 14 L/g to mL/kg **f** \$3600/g to c/mg

1G **13** If 20 metres of curtain material costs \$580, what would be the cost of 35 metres of the same material?

1G **14** A motor bike travels at a speed of 100 km/h. How far does it travel in 3.5 hours?

1G **15** A 5 kg bag of rice costs \$9.20. What is the cost of the following amounts?

a 10 kg **b** 40 kg **c** 3 kg
d 7 kg **e** 500 kg **f** 250 kg

1J **16** A car travels 960 km on 75 litres of petrol. How far does it travel on 50 litres?

Extended-response questions

1C **1** There are 384 passengers and 144 crew on a cruise ship. If there are 22 lifeboats, enough for everyone, how many are allotted to the passengers and how many to the crew?

1G **2** Pump A can fill a tank in 5 minutes; pump B can fill the tank in 10 minutes. How long will it take to fill the tank if both pumps are working together?

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